

Supplementary Material for

Study on CaO-promoted Co/AC catalysts for synthesis of higher

alcohols from syngas

Hong Du,^{a,c} Hejun Zhu,^{a*} Xingkun Chen,^{a,c} Wenda Dong,^a Wei Lu,^a Wenting Luo,^{a,c}
Miao Jiang,^{a,c} Tao Liu,^a Yunjie Ding^{a,b*}

^a Dalian National Laboratory for Clean Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, 116023, China

^b State Key Laboratory of catalysis, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, 116023, China

^c University of Chinese Academy of Sciences, Beijing 100039, China

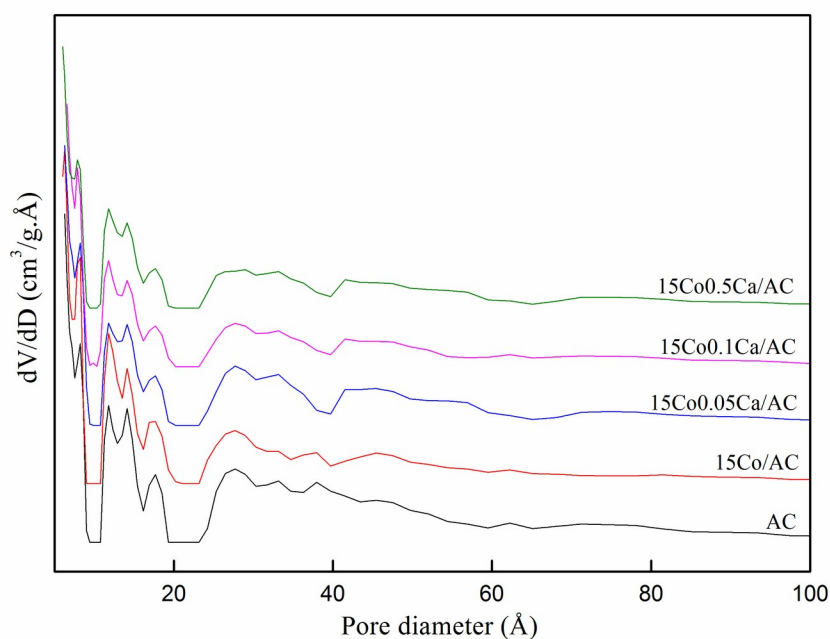


Figure S1. Pore size distribution curves of the AC support and corresponding catalysts calculated from non-local density functional theory (NLDFT).

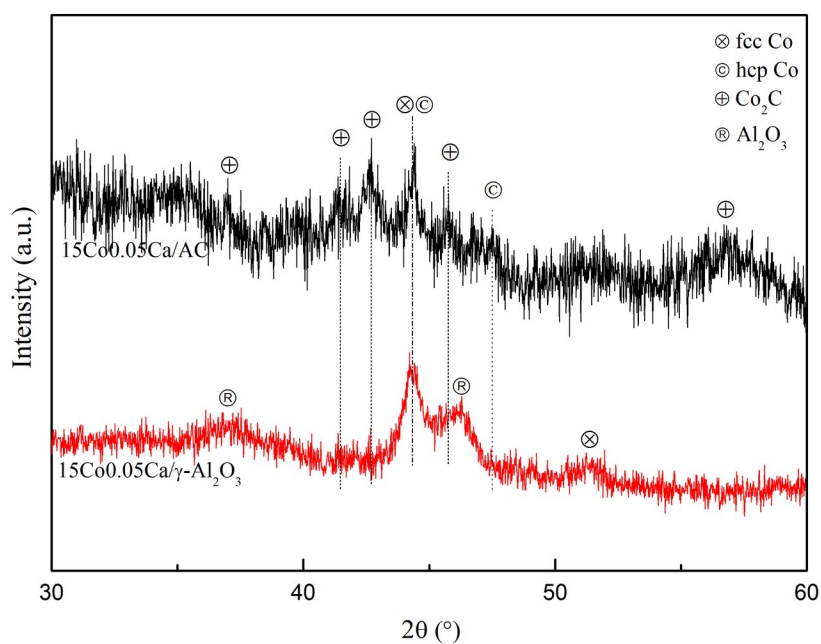


Figure S2. XRD patterns for the spent 15Co0.5Ca/AC and 15Co0.5Ca/γ-Al₂O₃ catalysts.

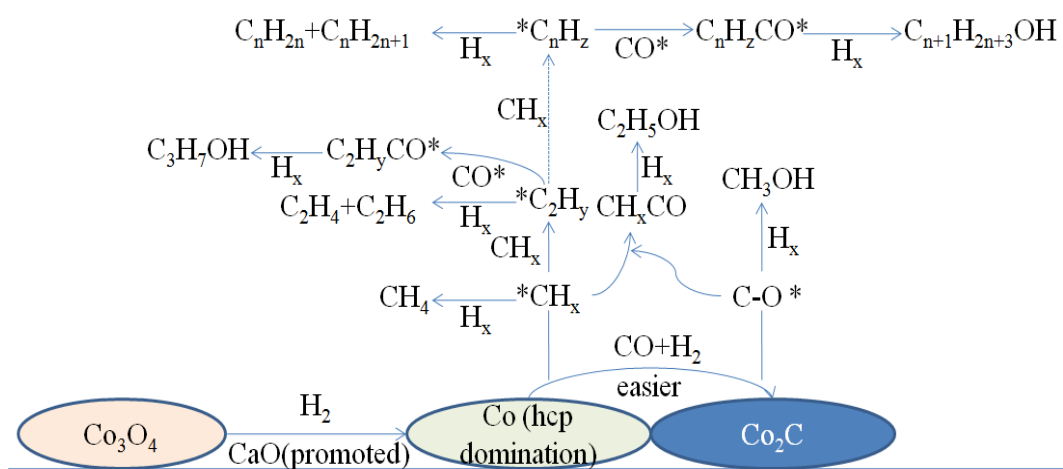


Figure S3. Scheme of CO hydrogenation reaction pathway over the 15Co-xCa/AC catalysts.

Table S1 The performance of CO hydrogenation over the 15Co0.5Ca/AC and 15Co0.5Ca/ γ -Al₂O₃ catalysts ^a

Catalyst	CO Conversion /%	Selectivity/C %				Alcohol Yield, %	Alcohol distribution/wt %	
		HC ₁₋₄ _b	CO ₂	HC ₅₊ _c	ROH _d		C ₁ OH ^e	C ₂₊ OH ^f
15Co0.5Ca/AC	24.7	48.0	1.3	21.4	29.3	7.2	4.2	95.8
15Co0.5Ca/ γ -Al ₂ O ₃	16.5	56.9	0.6	34.1	8.4	1.4	6.2	93.8

^a Reaction conditions: P = 3.0 MPa, T = 493 K, GHSV = 900 h⁻¹.

^b Hydrocarbons with carbon numbers of 1-4.

^c Hydrocarbons with carbon numbers more than 4.

^d Alcohols with carbon numbers of 1-18.

^e CH₃OH

^f Alcohols with carbon numbers more than one..